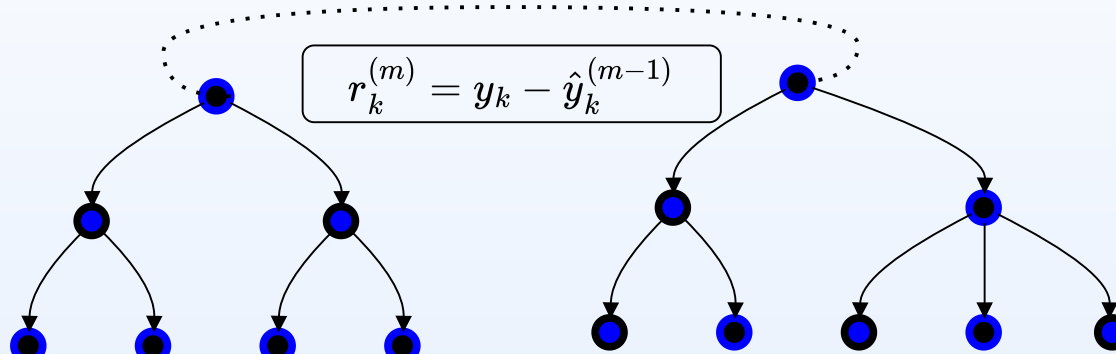


LightGBM Regressor (Gradient Boosting)

Boosting Iterations (m): 1, 2, ... , M



$$\hat{y}_k = \sum_{m=1}^M \eta f_m(x_k)$$

learning_rate (η)

Scales the contribution
of each tree

n_estimators (M)

Number of boosting
iterations (trees)

num_leaves

Controls number of leaf
nodes (regions)
Tree complexity

max_depth

Limits depth of the tree
(restricts recursive splits)

min_child_samples

Minimum samples per
leaf
(prevents small partitions)

Training Data
(X_{train}, y_{train})

Input features x_k
 $x_k = [I_{k-w}, \dots, I_{k-1}]$
(sliding window)

Validation Data
(X_{val}, y_{val})

Test Data
(X_{test}, y_{test})